Rational Functions: Full Practice Exam

Name: _____ Date: _____

13 questions | 58 points | 60 minutes

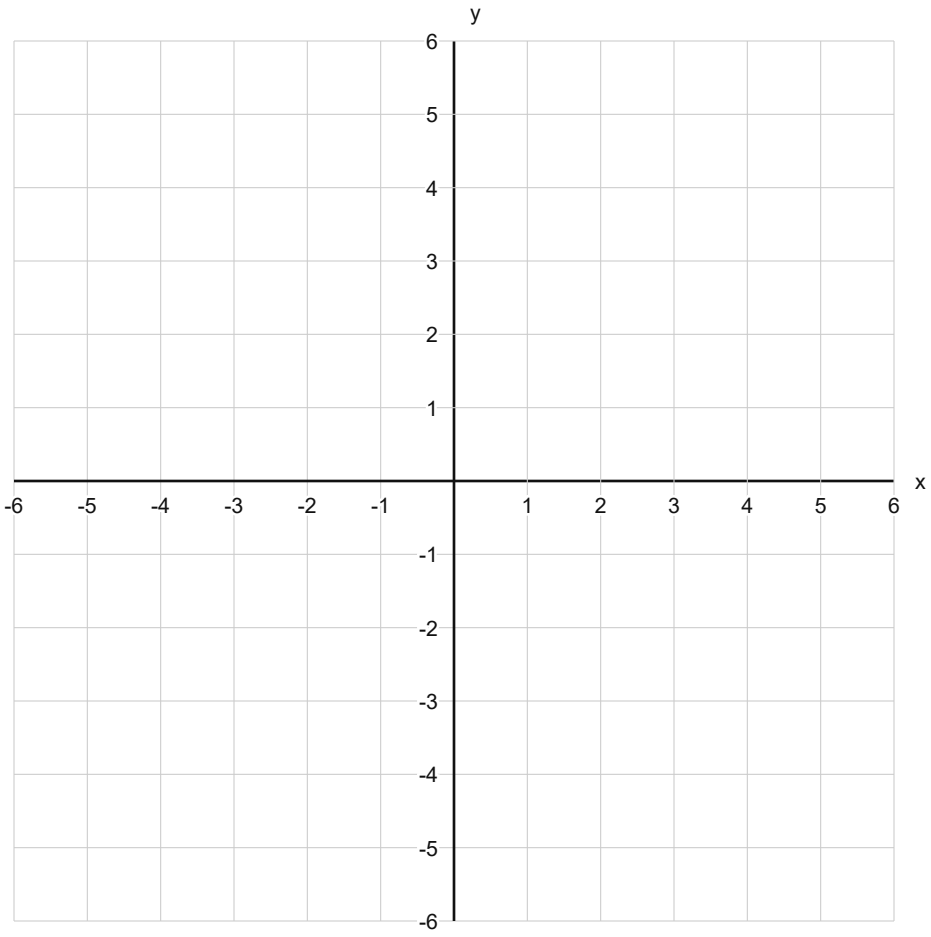
Complete every labeled part. Show the work and explanations requested. Give exact answers unless an estimate is requested. A four-function calculator is permitted.

---

1. 8 points

Let $f(x) = 3 - \dfrac{2}{x+1}$.

(a) Graph the function. Label its vertical and horizontal asymptotes and at least two points on the curve.



(b) State the far-left and far-right end behavior.

Rational Functions: Full Practice Exam

2. 4 points

Simplify $\frac{3}{x+2} - \frac{2x-1}{x^2-4}$. State all excluded real inputs. Show your algebra.

Work and final answer

3. 4 points

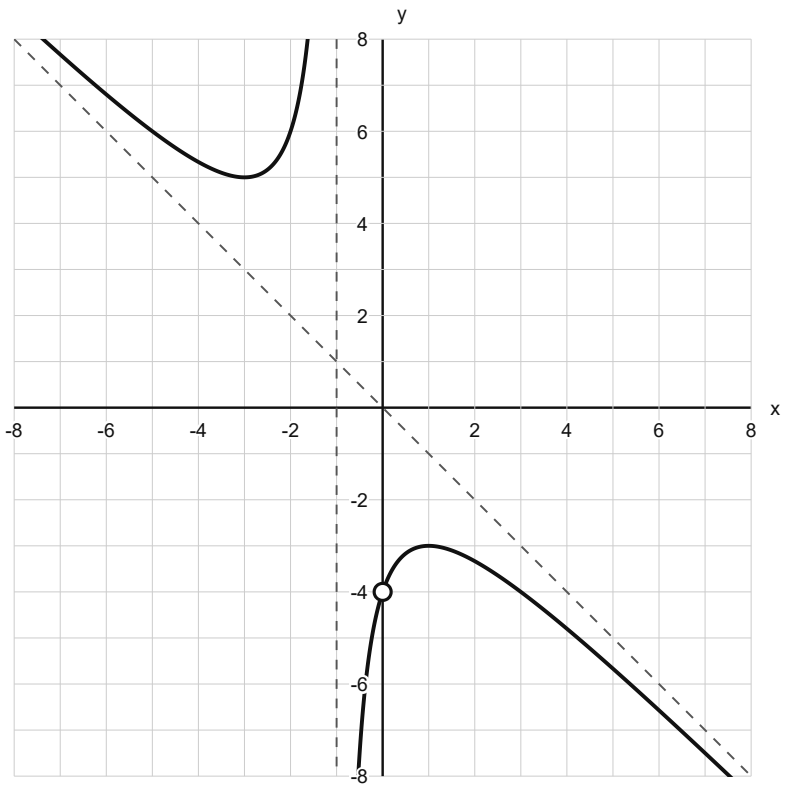
Simplify $\frac{x^2-2x-15}{x^2-25} \cdot \frac{x+5}{x^2-9}$. State all excluded real inputs. Show your algebra.

Work and final answer

Rational Functions: Full Practice Exam

4. 8 points

Use the supplied graph to answer each part. The curve continues beyond the window; an open circle denotes a missing point.



(a) Give the intervals where the function is increasing.

(b) Give the coordinates of every relative maximum and relative minimum, identifying each type.

(c) Give the intervals where the function is concave up and concave down.

(d) Identify all vertical, horizontal or slant asymptotes. Give their equations and types.

Rational Functions: Full Practice Exam

5. **10 points**

Let $R(x) = \dfrac{2x^2 + 6x + 4}{x^2 - x - 2}$.

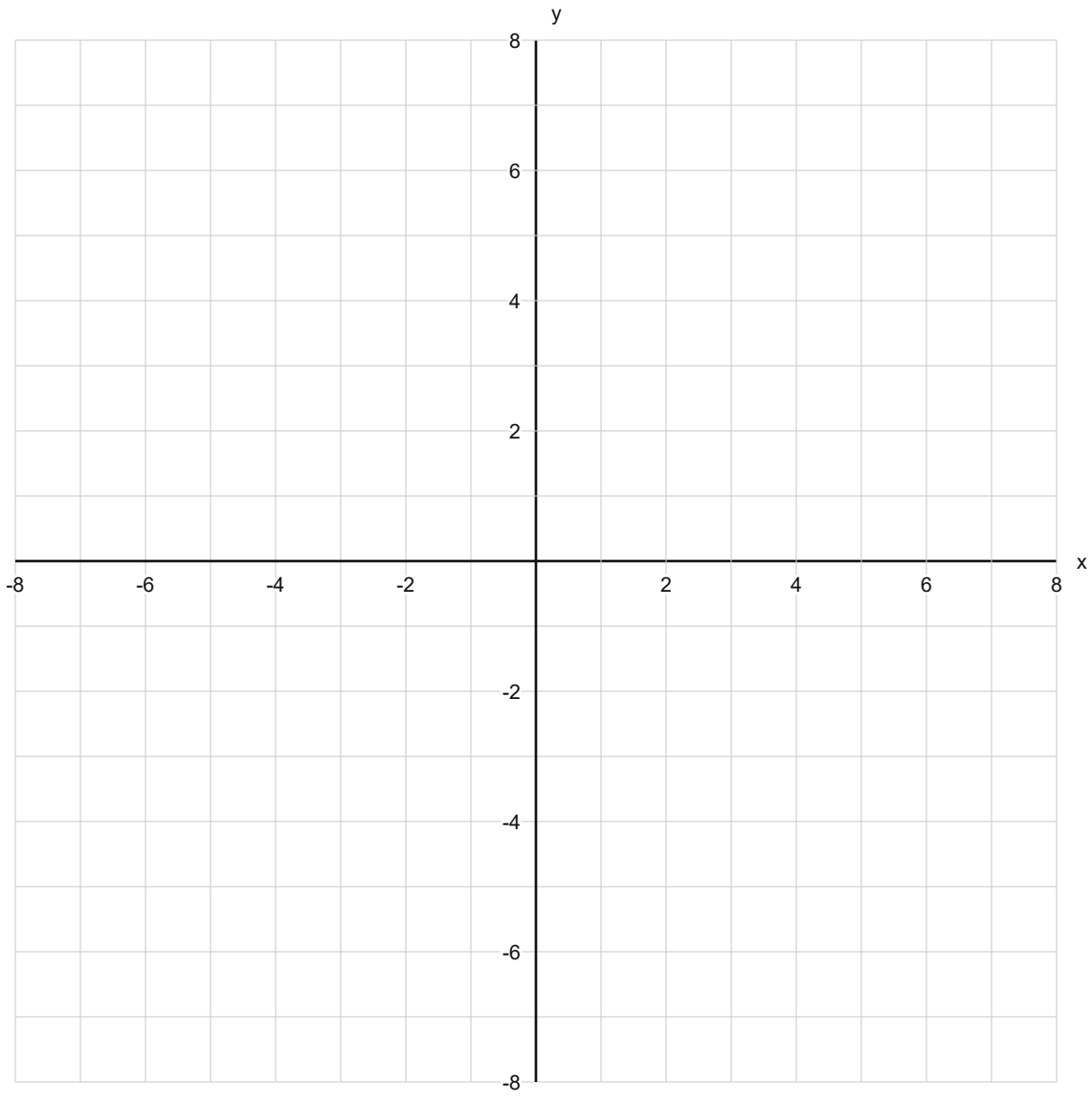
(a) Find all intercepts, holes and asymptotes. Give exact coordinates or equations as appropriate and show your algebra.

Continue with part (b) on the next page.

Rational Functions: Full Practice Exam

5. (continued)

(b) Sketch the function on the separate grid page and label the features you found.



Rational Functions: Full Practice Exam

6. 4 points

Let $S(x) = \dfrac{-x^2 + 5}{2x - 4}$. Find all horizontal or slant asymptotes and state the far-left and far-right end behavior. Justify the asymptote algebraically.

(a) Algebra and asymptote conclusion

(b) Far-left and far-right behavior, with an algebraic explanation

7. 4 points

A workshop has made 10 kits at a total cost of \$180. Each additional kit costs \$12 to make. After n additional kits, the average cost per kit is $C(n) = \dfrac{12n + 180}{n + 10}$, for nonnegative integers n. Interpret the long-term average cost, including units. State whether it approaches that value from above or below and whether it reaches the value for any finite n. Justify your conclusions.

Answer and justification

Rational Functions: Full Practice Exam

8. **4 points**

Construct a rational function whose only x-intercept is $(-4, 0)$, whose only vertical asymptote is $x = -1$, whose horizontal asymptote is $y = 3/2$, and whose only hole is at $x = 2$.

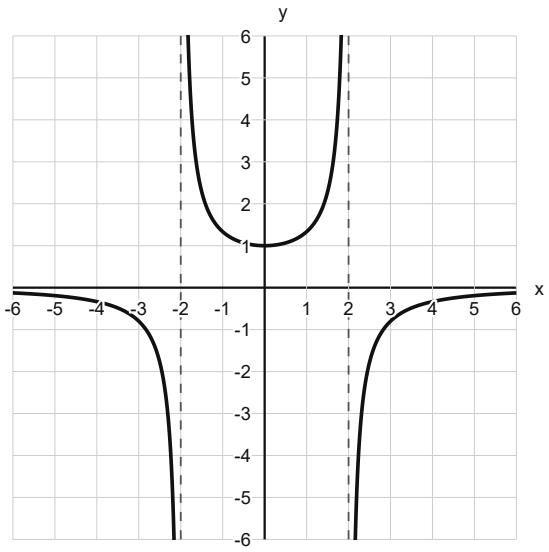
(a) Write your function in factored form and verify that it has each required feature.

(b) Give the coordinates of the hole for your function. Show your calculation.

Rational Functions: Full Practice Exam

9. 2 points

The graph represents a nonzero rational function. Which statement about its numerator and denominator degrees must be true? Select one answer and justify your choice.



A. The degrees are equal.

B. The denominator degree is greater than the numerator degree.

C. The numerator degree is exactly one greater than the denominator degree.

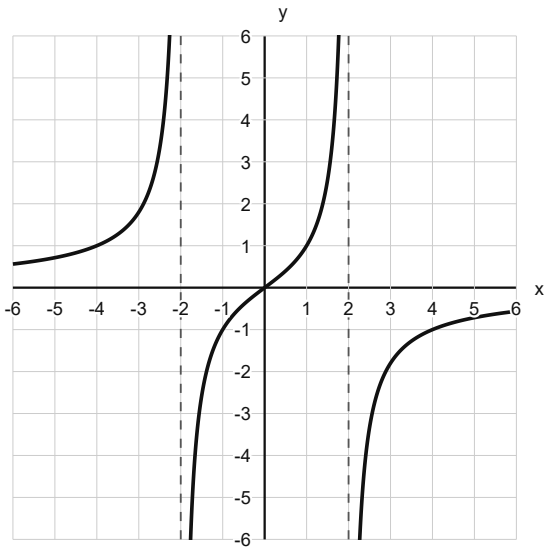
D. The numerator degree is exactly two greater than the denominator degree.

Choice and justification

Rational Functions: Full Practice Exam

10. **2 points**

Which equation matches the graph? Select one answer and justify your choice.



A. $y = \dfrac{-3x^2}{x^2 - 4}$

B. $y = \dfrac{3x}{x^2 - 4}$

C. $y = \dfrac{-3x}{x^2 + 4}$

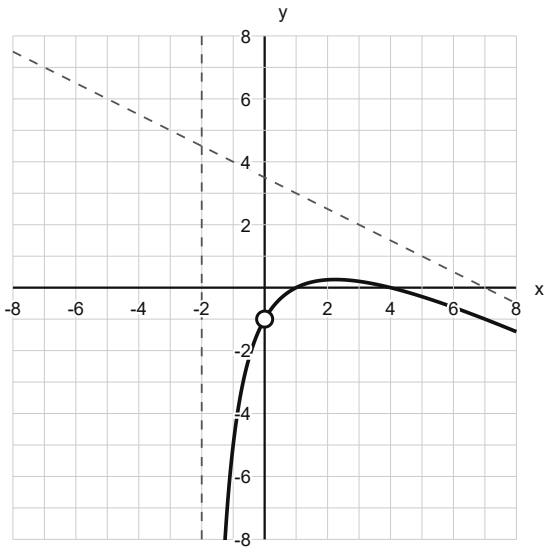
D. $y = \dfrac{-3x}{x^2 - 4}$

Choice and justification

Rational Functions: Full Practice Exam

11. 2 points

Which function matches the entire graph, including any missing point? Select one answer and justify your choice.



A. $y = \dfrac{-x(x-1)(x-4)}{2x(x+2)}$

B. $y = \dfrac{-(x-1)(x-4)}{2(x+2)}$

C. $y = \dfrac{-x(x-1)(x-4)}{2x(x-2)}$

D. $y = \dfrac{x(x-1)(x-4)}{2x(x+2)}$

Choice and justification

Rational Functions: Full Practice Exam

12. 2 points

Let $f(x) = -5x^6 + 2x - 1$ and $g(x) = \dfrac{3x^m - 8x^4 + 2}{x^2 + 1}$, where m is a positive integer. Which value cannot make the far-left and far-right end behavior of g match that of f? Select one answer and justify your choice.

A. $m = 1$

B. $m = 4$

C. $m = 5$

D. $m = 3$

Choice and justification

13. 4 points

A rational function H has only two undefined real inputs, 1 and 3. One is a hole and the other is a vertical asymptote. Selected values are shown; decimals are rounded. Identify the x-coordinate of each and explain the evidence in the table for each classification.

x	0.9	0.99	0.999	1	1.001	1.01	1.1
H(x)	-17	-197	-1997	undefined	2003	203	23

x	2.9	2.99	3	3.01	3.1
H(x)	4.0526	4.005	undefined	3.995	3.9524

(a) Hole location and supporting table evidence

(b) Vertical-asymptote location and supporting table evidence